



NITI Aayog

SECTOR AT A GLANCE 05

Green Hydrogen Energy in India

Energy in India

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SECTOR AT A GLANCE 05

GREEN HYDROGEN ENERGY IN INDIA

ENERGY IN INDIA

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Green hydrogen is produced using renewable energy sources like solar, wind, or biomass. Currently, India annually consumes about 6.8 million metric tonnes (MMT) of hydrogen.¹ Most of this is grey, meaning it is produced using fossil fuels such as natural gas and naphtha. The primary driver for this demand is the industrial sector, majorly entailing hydrogen use for petroleum refining and producing fertilisers, methanol, and metals.²

Green hydrogen holds significant promise in facilitating low-carbon and self-sufficient economic pathways for India. India has massive endowments of renewables like solar and wind energy, highlighting the huge potential for the growth of green hydrogen. In the run-up to the 18th G20 Summit held in India, a conference on Green Hydrogen Pilots in India was held that showcased various innovative GH2 pilots that India is pioneering.³ The G20 New Delhi Leaders Declaration also stressed the importance of accelerating the production, use, and development of a global market for GH2.⁴

However, GH2 production and adoption in India face certain challenges, such as high cost of production, limited end-use cases, and limited investments in infrastructure development at scale.

Given these opportunities and challenges, enhancing the production, use and export of GH2 are key policy priorities. Several ongoing and recent central government initiatives, including but not limited to the National Green Hydrogen Mission, focus on expanding GH2 production and adoption in India. Efforts under this mission are expected to generate approximately six lakh jobs by 2030.⁵

Accelerating domestic manufacturing, promoting skill development, and encouraging greater adoption across end-use sectors and industries will be critical to advancing India's green hydrogen ambitions. This note presents key trends related to different components of green hydrogen in India and actions for the way forward.

Figure 1: Key green hydrogen energy statistics

6.8 MMT

Annual hydrogen consumption
in India

INR 19,744 crores

Financial outlay for the National Green
Hydrogen Mission (NGHM)

INR 1 lakh crore

Estimated import savings if NGHM
outcomes are achieved

USD 8 billion

Estimated value of the green
hydrogen market in India by
2030

3.6 giga tonnes

Estimated cumulative CO2 emissions
reductions by 2050 due to GH2 adoption

USD 5 billion

Estimated size of the electrolyser
market in India by 2030

Source: NITI Aayog, Harnessing Green Hydrogen, 2022; Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023; Ministry of New and Renewable Energy, National Hydrogen Mission – An Update, 2023;

¹ NITI Aayog, Harnessing Green Hydrogen, 2022; Department of Science and Technology, India Country Status Report on Hydrogen and Fuel Cells, 2020

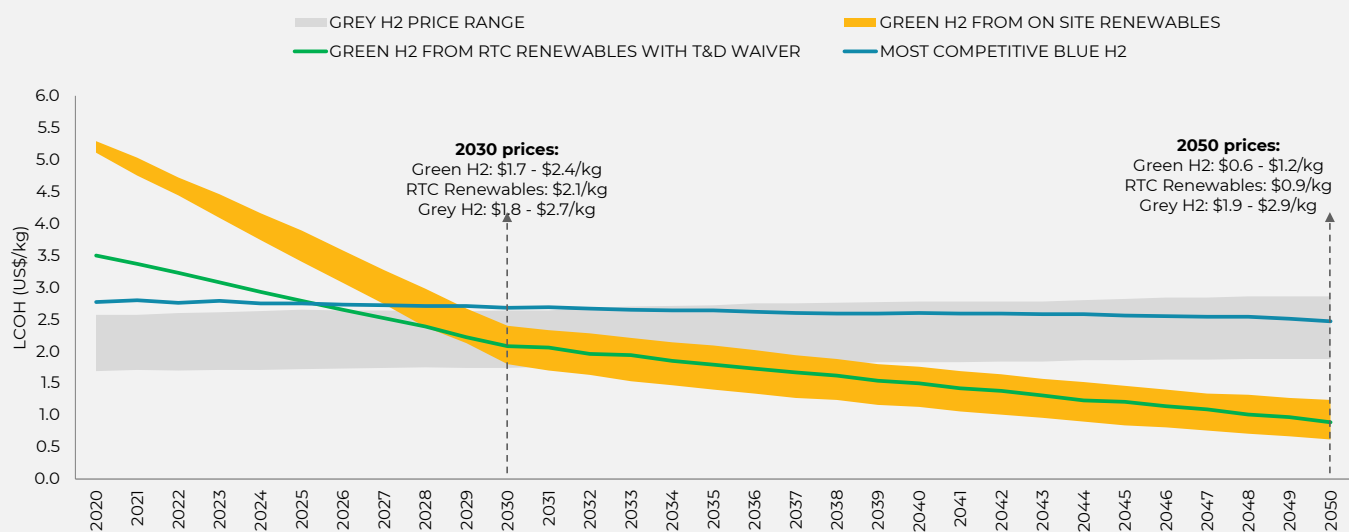
² Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

³ Ministry of New and Renewable Energy, 'Green Hydrogen Pilots in India' Conference held in the run-up to G20 Summit, 2023

⁴ Ministry of External Affairs, G20 New Delhi Leaders' Declaration, 2023

⁵ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

Figure 2: Projected price trajectory of solar-green hydrogen production based on decline in electrolyser and renewable costs



Source: IEA, BNEF, TERI, SECI, and RMI analysis in NITI Aayog, Harnessing Green Hydrogen, 2022
 Note: Currency conversion: \$1 = INR 72

Key trends

This section dives deeper into India's green hydrogen energy landscape, outlining statistics and trends on:

1. Resource planning and data management
2. Hydrogen production and storage
3. Hydrogen distribution
4. Hydrogen utilisation
5. Workforce development
6. Waste management
7. Research and development
8. International collaboration
9. Monitoring and evaluation

Resource planning and data management

Green hydrogen requires large quantities of water and renewable energy, such as solar, wind and biomass, to power electrolyzers.

India aims to be a global hub for green hydrogen production, usage, and export:

To achieve this, India plans to capitalise on its abundant renewable energy sources, implement favourable policies, make strategic investments, and foster international partnerships. The total financial layout for the mission between 2023-2030 is INR 19,744 crores, of which 87% is allocated for the Strategic Interventions for Green Hydrogen Transition (SIGHT) program, 7% for pilot projects, remaining 6% for research and development (R&D) and other miscellaneous components of the mission.⁶

By 2050, hydrogen is expected replace significant amounts of imported fossil fuels:

Industries like ammonia production, refineries, and methanol production, are expected to transition from natural gas to green hydrogen, leading to the displacement of 35 million tonnes of oil equivalent (Mtoe) of natural gas.⁷ According to the Ministry of New and Renewable Energy, an increase in GH2 capacity by 2030 can reduce fossil fuel imports of over INR 1 lakh crore.⁸ Significant amounts of CO2 emissions are produced by heavy-duty trucking, aviation, and maritime shipping, and hydrogen is being explored as one of the primary alternatives for these sectors.⁹ While some natural gas usage in transportation is estimated to be replaced by hydrogen, this impact is relatively minimal due to the presence of electric vehicles (EVs) and the limited role of natural gas as a transport fuel.¹⁰

India needs to rapidly advance its power-H2-power capabilities to contribute 125 GW of GH2 to its renewable energy mix by 2030:

The production of hydrogen through electrolysis requires a substantial amount of electricity and involves significant energy losses during the conversion process. GH2 becomes cost-effective to meet power needs only under specific circumstances when prices align with the projected rates for the year 2050.¹¹

⁶ Ministry of New and Renewable Energy, Cabinet approves National Green Hydrogen Mission, 2023

⁷ The Energy and Resources Institute, The Potential Role of Hydrogen in India: A pathway for scaling-up low carbon hydrogen across the economy, 2020

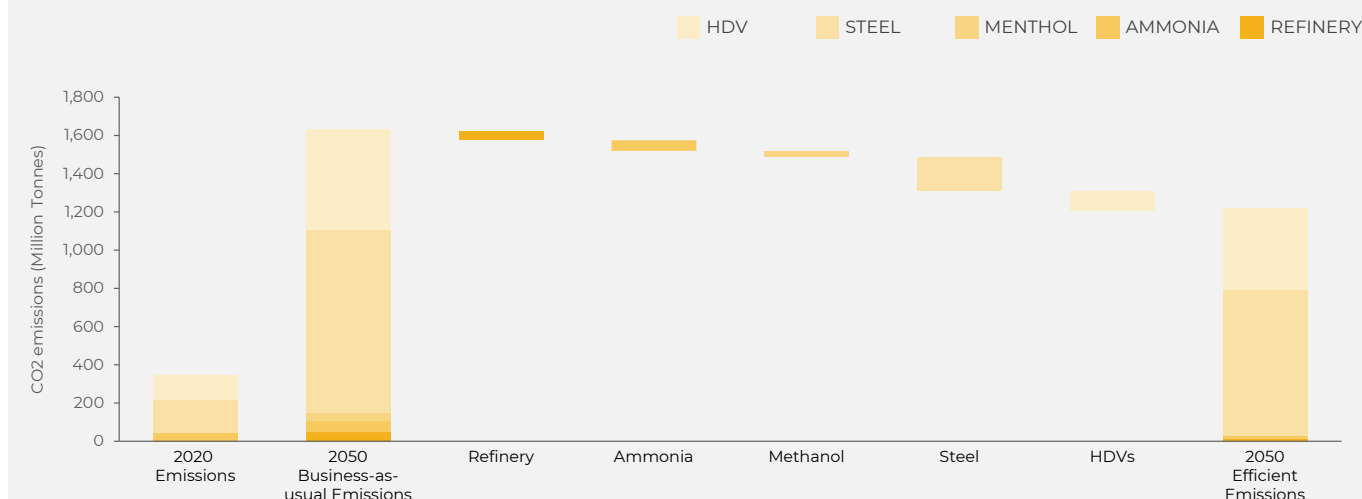
⁸ National portal of India, National Green Hydrogen Mission, accessed July 2023

⁹ NITI Aayog, Harnessing Green Hydrogen, 2022

¹⁰ The Energy and Resources Institute, The Potential Role of Hydrogen in India: A pathway for scaling-up low carbon hydrogen across the economy, 2020

¹¹ NITI Aayog, Harnessing Green Hydrogen, 2022

Figure 3: CO2 emissions reductions in 2050 due to green hydrogen uptake



Hydrogen production and storage

India will need to ramp up its domestic production capabilities to meet its targets.

As of 2023, most of India's hydrogen is obtained via fossil fuels:

Natural gas is the primary energy source for hydrogen production. In the past, a more significant portion of hydrogen was generated from naphtha, but this has been reduced due to the widespread adoption of natural gas in the fertiliser industry. In recent years there has been renewed interest in coal gasification to decrease energy costs and reduce dependence on energy imports. However, coal gasification is an emissions-intensive process.¹²

India has set a target of annual production of 5 MMT of green hydrogen by 2030: India aims to contribute about 125 GW of renewable energy capacity via green hydrogen by 2030.¹³ To accomplish this objective, the National Green Hydrogen Mission emphasises advancing research, development, and implementation of green hydrogen technologies. It also aims to establish a comprehensive ecosystem for producing, storing, and utilising green hydrogen.¹⁴ India's first pure green hydrogen plant was established at the Jorhat Pump Station in Assam in 2022. It draws power from a 500kW solar plant and can produce pure green hydrogen. However, it currently only has a daily production capacity of 10 kg.¹⁵

India is building incentive schemes to boost electrolyser manufacturing capabilities:

The Ministry of New & Renewable Energy announced in June 2023 that the incentive scheme for electrolyser manufacturing and green hydrogen production has been drafted and will be implemented soon. The total incentives provided under this initiative will exceed INR 17,000 crores until 2030 and will be released in phases.¹⁶

MNRE has recently notified the GH2 standard for India:

The Ministry of New and Renewable Energy, in August 2023, released the Green Hydrogen Standard for India, outlining the definition of green hydrogen in India and providing minimum standards for hydrogen

¹² The Energy and Resources Institute, The Potential Role of Hydrogen in India: A pathway for scaling-up low carbon hydrogen across the economy, 2020

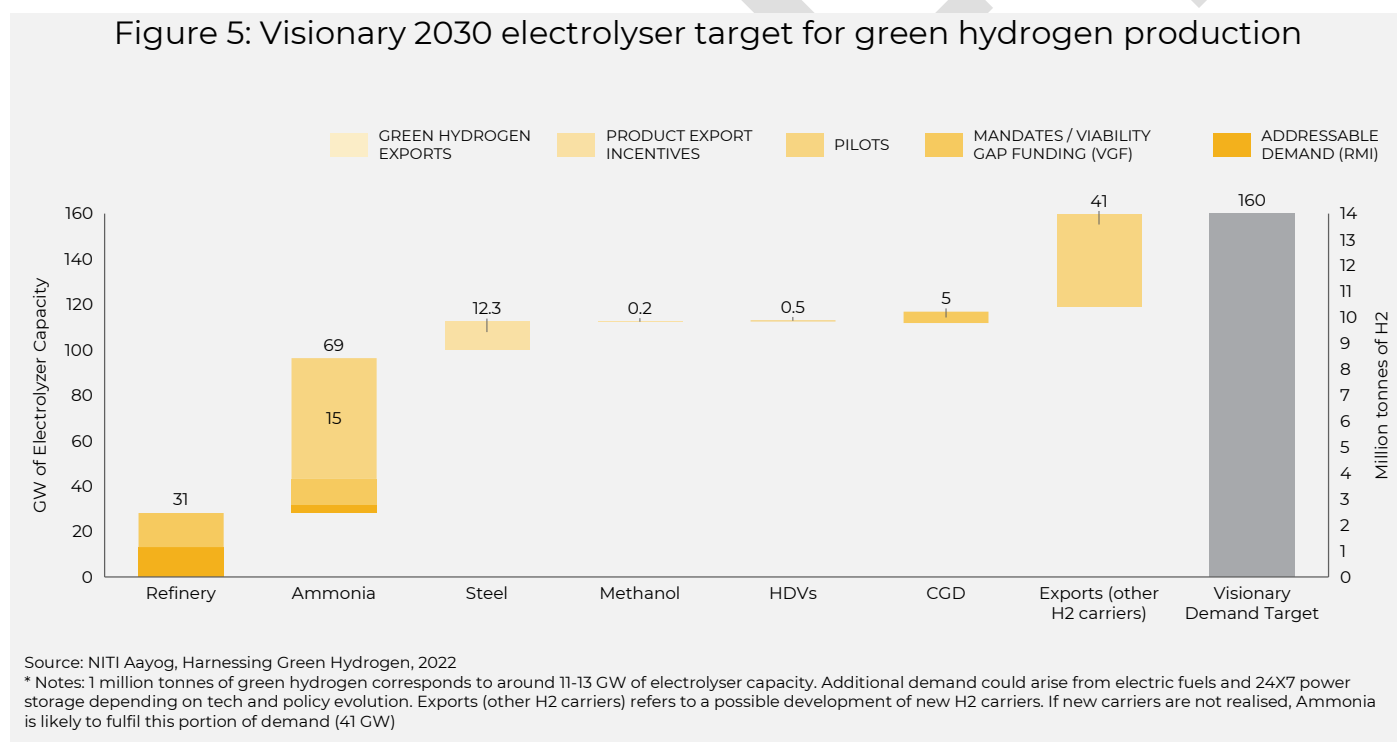
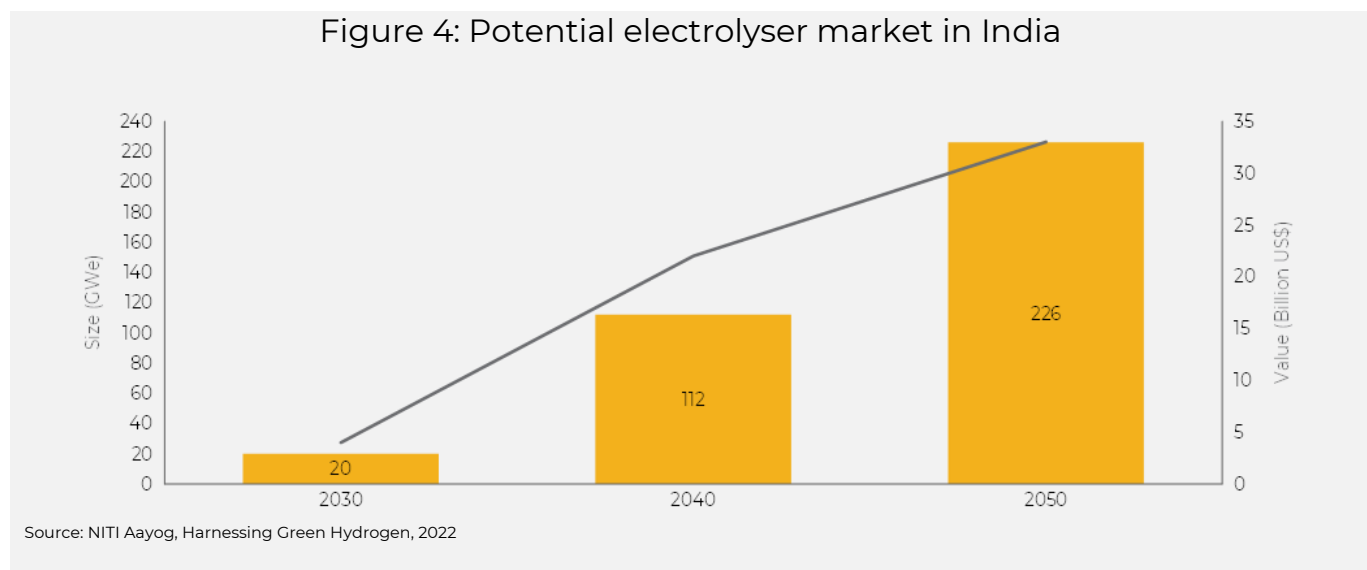
¹³ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

¹⁴ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

¹⁵ Ministry of Petroleum & Natural Gas, India's first pure green hydrogen plant commissioned in Jorhat, 2022

¹⁶ Ministry of New and Renewable Energy, International Conference on Green Hydrogen to be held in New Delhi during July 5-7, 2023

production via electrolyzers or biomass.¹⁷ It also provides detailed of the nodal agency for all GH2 projects in India, the Bureau of Energy Efficiency under the Ministry of Power.



Hydrogen distribution

Currently, hydrogen produced in India is mainly for captive usage. Given the growth of this sector, significant investments are required to build a dedicated hydrogen distribution ecosystem.

Currently, India has limited hydrogen transportation needs:

Hydrogen can be transported as a compressed gas through pipelines or trucks/tankers, with the former preferred over long distances. Hydrogen gas can also be condensed to a liquid state or can be transported through Liquid Organic Hydrogen Carriers (LOHCs), which are chemical compounds that store and

¹⁷ Ministry of New and Renewable Energy, Green Hydrogen Standard for India, 2023

release hydrogen through reversible chemical reactions. The main disadvantage of transporting liquid hydrogen is "boil-off," which refers to the loss of hydrogen if the low temperatures required for its storage are not adequately maintained. Other chemical carriers which can be used for easy transport of hydrogen include ammonia or methanol. However, the final consumption of hydrogen necessitates further processing, which in turn requires extra energy use and associated costs for such conversation.¹⁸ Most hydrogen in India is produced and consumed on the same facility by specific industries, resulting in negligible transportation needs at its current scale. For instance, in refineries, which are one of the largest hydrogen consumers in India, most of the hydrogen is produced and consumed on-site.¹⁹

Trucks and pipeline retrofitting are suited to India's short-term hydrogen distribution requirements:

The use of compressed hydrogen trucks is most appropriate for India's nascent hydrogen market given suitability for smaller volumes and because it is more economically viable than establishing pipelines. Further, India is expected to retrofit its existing natural gas pipeline network to accommodate hydrogen.

Hydrogen utilisation

Presently, Indian industries consume the majority of the grey hydrogen produced. Future demand projections show a positive outlook for green hydrogen.

The industrial sector is the largest consumer of hydrogen in India:

As per a NITI Aayog report dated 2022, India's current hydrogen consumption stands at 6 million tonnes.²⁰ As mentioned above, all of this is grey hydrogen. Majority of the hydrogen produced is used for industrial purposes like oil refining, which accounts for 33% of total use, and ammonia production which is at 27%, as per 2020 figures. A smaller share of hydrogen use is attributed to methanol production which stands at 11%, followed by steel production at 3%.²¹

Demand for hydrogen is expected to reach approximately 30 million tonnes by 2050:

Hydrogen demand has the potential to increase more than four times from 2020 to 2050, of which 94% could be met by green hydrogen. In the short term, industrial feedstock demand (ammonia and refining markets) can touch 11 million tonnes per annum by 2030, an ~83% increase from current numbers.²²

Steel and transport sectors are expected to comprise a significant share of hydrogen demand in the future:

In addition to its existing industrial use cases, green hydrogen can decarbonise hard-to-abate sectors like steel and long-haul transport. These sectors are projected to drive long-term hydrogen demand, where steel and heavy-duty trucking are expected to account for 52% of total demand by 2050. The National Green Hydrogen Mission also prescribes conducting pilot projects in steel production, long-haul heavy-duty mobility, and shipping. Aspirational targets set by the government include reduction of carbon intensity of old steel plants to less than 2 tonnes of CO₂ per tonne of steel, by 2035. With respect to heavy duty vehicles, by 2030, it aims to pilot 1,000 trucks, 50 boats, and 10 aircrafts. Additionally, three hydrogen corridors are set to be developed in India.²³

Significant efforts are already underway for building a standards and regulatory framework for the hydrogen ecosystem:

¹⁸ The Energy and Resources Institute, The Potential Role of Hydrogen in India: A pathway for scaling-up low carbon hydrogen across the economy, 2020; NITI Aayog, Harnessing Green Hydrogen, 2022

¹⁹ NITI Aayog, Harnessing Green Hydrogen, 2022; The Energy and Resources Institute, The Potential Role of Hydrogen in India: A pathway for scaling-up low carbon hydrogen across the economy, 2020; South Asia Regional Energy Partnership, Investment Landscape of Green Hydrogen in India, 2023

²⁰ Note: Given that India's current production of hydrogen is 5 MMT, this figure assumes that India imports about 1 million tonnes of hydrogen

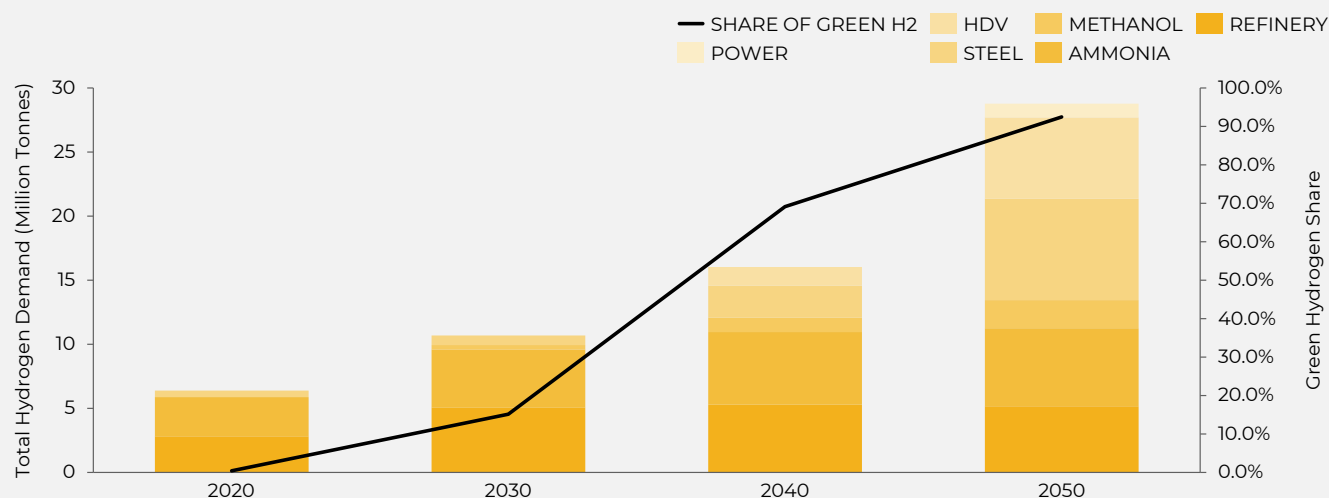
²¹ NITI Aayog, Harnessing Green Hydrogen, 2022; Department of Science and Technology, India Country Status Report on Hydrogen and Fuel Cells, 2020

²² NITI Aayog, Harnessing Green Hydrogen, 2022

²³ NITI Aayog, Harnessing Green Hydrogen, 2022; Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

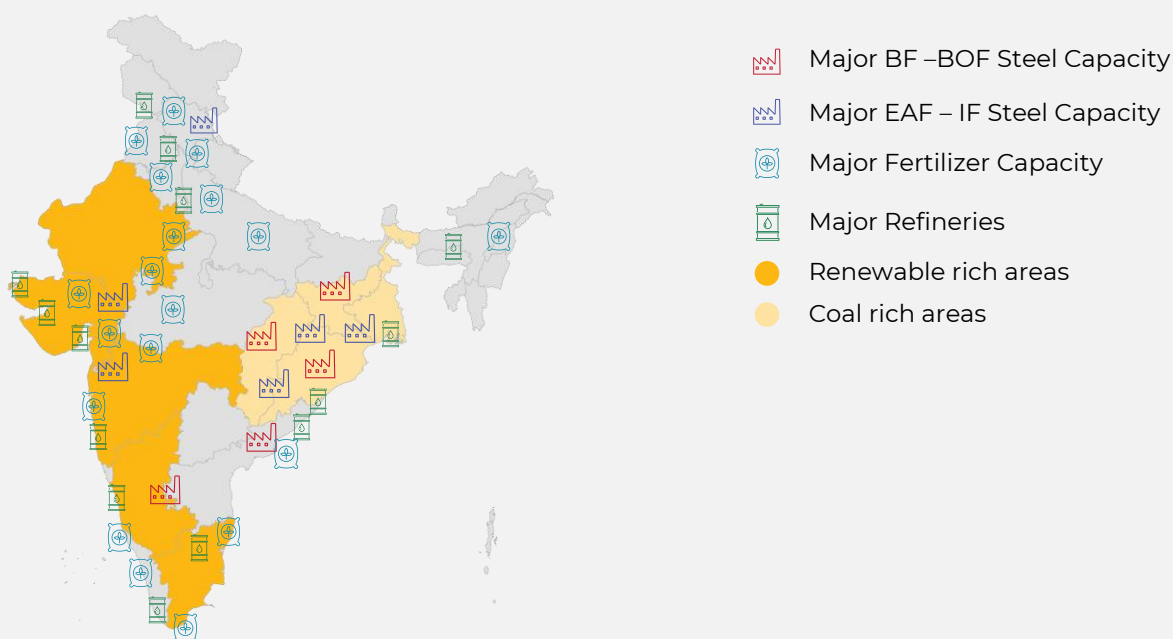
The Bureau of Indian Standards is actively formulating and issuing standards to cover various critical aspects, focusing on adopting relevant international standards whenever possible. Additionally, the Ministry of Road Transport and Highways is working on developing specific standards and regulations for automotive applications of green hydrogen.²⁴

Figure 6: Hydrogen demand outlook and potential green hydrogen share at cost parity (without policy intervention)



Source: NITI Aayog, Harnessing Green Hydrogen, 2022

Figure 7: Location of key industrial facilities



Source: NITI Aayog, Harnessing Green Hydrogen, 2022

²⁴ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

Workforce development

India's energy workforce needs specialised skills and reskilling efforts to match the future growth potential of green hydrogen in the economy.

The green hydrogen sector can generate 6 lakh jobs by 2030:

Across the green hydrogen value chain, the supply of renewable energy such as solar and wind to produce green hydrogen is expected to create over 66% of new jobs. Plant construction and green hydrogen production will account for the second largest share of jobs, at 22% collectively.²⁵ Electrolyser manufacturing and distribution segment will account for 4% each, and 2% will be attributed to green hydrogen storage.²⁶

The National Green Hydrogen Mission has provisions for skill-building:

The mission outlines that the Ministry of Skill Development and Entrepreneurship (MSDE) will work with MNRE and other ministries to develop skills needed for employment in the GH2 sector. The MSDE has formed a committee with the Skills Council for Green Jobs to create courses and programs to train students in various aspects of the value chain such as power electronics, electrolyzers, fuel cells, advanced materials, hydrogen storage, covering design, manufacturing and installation, compression and distribution, and operations and maintenance.²⁷ The Ministry of Education will work to ensure that hydrogen technologies and the latest developments are included in the educational curriculum at different levels. They will encourage practical experience with these technologies through guidelines for setting up laboratories in schools and higher education institutions. A major part of skilling initiatives will focus on reskilling the existing workforce in sectors which are harmful to the environment, such as coal and petroleum, and absorbing them into the GH2 ecosystem.²⁸

Waste management

India's transition to green hydrogen supported by targeted programs like the National Green Hydrogen Mission assume importance given the current environmental pollution caused by grey hydrogen consumption.

Green hydrogen production has no associated harmful by-products:

Green hydrogen is generated by using renewable energy to operate an electrolyser, which separates hydrogen from water molecules. This process produces no associated pollutant or harmful by-product. On the other hand, other forms of hydrogen like grey or black, release detrimental greenhouse gases like carbon dioxide.²⁹ In the case of coal-based hydrogen production, the emissions can exceed 600 gCO₂/kWh hydrogen produced.³⁰

Research & Development

India's green hydrogen R&D efforts are guided by the R&D Roadmap for Green Hydrogen Ecosystem in India 2023.

India's current GH2 R&D focus is on material, technology, and infrastructure development to make GH2 production, storage, and transportation viable and cost-effective:

²⁵ Skill Council for Green Jobs, Gearing up the Indian workforce for a Green Economy, 2023

²⁶ Skill Council for Green Jobs, Gearing up the Indian workforce for a Green Economy, 2023

²⁷ Skill Council for Green Jobs, Gearing up the Indian workforce for a Green Economy, 2023

²⁸ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

²⁹ National grid, What is hydrogen?, accessed July 2023

³⁰ The Energy and Resources Institute, The Potential Role of Hydrogen in India: A pathway for scaling-up low carbon hydrogen across the economy, 2020

India's current R&D efforts in hydrogen aim to reduce the high capital and operational costs of electrolysis, making hydrogen production more cost-effective. The focus is on lowering electrolyser expenses and enhancing their operational efficiency and durability. To further decrease the carbon footprint, initiatives include reintegrating electrolysers into the power system to support higher renewable energy concentrations, ensuring material circularity, and developing large-scale electrolyser systems. Supported by favourable policy measures, these advancements aim to produce renewable hydrogen at a cost below INR 100/kg, driving the country's green hydrogen ambitions forward.³¹

The government has a dedicated budget of about Rs. 400 crores for GH2 R&D:

The funding is under the National Green Hydrogen Mission and aims to make green hydrogen production, storage, distribution, and utilisation more affordable while improving the efficiency, safety, and reliability of related systems and processes by 2030.³² The government also plans to create a public-private partnership framework called the Strategic Hydrogen Innovation Partnership (SHIP) to support R&D projects.³³

International collaboration

India has established partnerships with leading green hydrogen economies and international alliances:

India signed a joint declaration of intent to formulate an Indo-German Hydrogen Task Force in 2022 to enhance collaborative efforts in the production, utilisation, storage, and distribution of green hydrogen. The partnership establishes frameworks facilitating GH2 projects, regulations, standards and joint research and development (R&D) initiatives.³⁴ India has also entered into a strategic agreement with the International Renewable Energy Agency (IRENA) to progress towards cost-effective decarbonisation via developing and deploying green hydrogen.³⁵

India aims to take a prominent role in international GH2 alliances for knowledge exchange and trade cooperation:

The National Green Hydrogen Mission highlights the Indian government's intention to form multilateral collaborations with alliances such as the International Partnership on Hydrogen and Fuel Cells in the Economy, Clean Energy Ministerial and Mission Innovation.³⁶ As highlighted by the NITI Aayog, partnerships with specific countries like Japan, Korea, and Germany can help India advance green hydrogen commercialisation. This involves collaborations involving knowledge exchanges on setting quality standards for green hydrogen production and promoting the export of Indian products such as green steel and ammonia.³⁷

Monitoring and evaluation

National efforts are underway to ensure a robust regulatory framework for the hydrogen ecosystem.

NGHM and states are looking to set up monitoring committees to ensure accountability towards green hydrogen targets:

The National Green Hydrogen Mission outlines the provision of an Empowered Group (EG), a national level working group comprising of representatives from NITI Aayog, key ministerial secretaries and industry

³¹ Ministry of New and Renewable Energy, R&D Roadmap for Green Hydrogen Ecosystem in India, 2023

³² Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023; Ministry of New and Renewable Energy, Cabinet approves National Green Hydrogen Mission, 2023

³³ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023; Ministry of New and Renewable Energy, Cabinet approves National Green Hydrogen Mission, 2023

³⁴ Ministry of New and Renewable Energy, Joint declaration of Intent on Indo – German Green Hydrogen Task Force signed, 2022

³⁵ Union Cabinet, Cabinet approves Strategic Partnership Agreement between the Ministry of New and Renewable Energy (MNRE), India and the International Renewable Energy Agency (IRENA), 2022

³⁶ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

³⁷ NITI Aayog, Harnessing Green Hydrogen, 2022

experts. The EG is tasked with overseeing NGHM activities, suggest policy interventions, and monitor progress and implementation.³⁸ Uttar Pradesh in 2022 became one of the first states to have released a draft green hydrogen policy. The policy mandates the formulation of a state-level committee to monitor overall progress, manage finances, approve incentives for manufacturers and other beneficiaries, revise targets as per ongoing progress, and release guidelines and standards for the production and storage of GH₂.

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³⁸ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by increasing the adoption of green hydrogen. These revolve primarily around the potential for decarbonisation of fossil-fuel based sectors, efficient energy storage, and increased domestic GH2 production.



Accelerate India's decarbonisation and RE transition:

By 2030, India aims to reach a green hydrogen production capacity of at least 5 MMT annum, and these efforts are expected to add about 125 GW of renewable energy capacity. In line with initiatives like the National Green Hydrogen Mission, approximately 50 MMT of CO2 emissions per annum are expected to be combatted.³⁹



Drive sustainable operations in the industrial and mobility sectors:

Green hydrogen can directly replace fossil fuel-based resources like fuel or industrial feedstock. It can substitute grey hydrogen in industrial use cases like petroleum refining and fertiliser production. Further, it can be used in long-haul vehicles and marine vessels, replacing unsustainable fuels like diesel.⁴⁰



Serve as a crucial energy storage solution:

Green hydrogen can be explored as an important and versatile energy carrier and means of storage. It can thus be used to meet the energy requirements of remote regions, including distant islands.⁴¹



Expand production of green hydrogen:

In India, most of the hydrogen presently consumed is grey in nature, i.e. sourced from fossil fuels. India has excellent potential for producing green hydrogen as it has vast resources of renewable energy and land. Further, the government has displayed strong support through initiatives like the National Green Hydrogen Mission, and pilot projects have also commenced. This positions India well to cater to domestic demand and become a global hydrogen exporter.⁴²



Establish India as a first mover and future leader in global GH2 markets:

The global demand for green hydrogen and its derivatives is expected to rise in line with the net zero revolution. India has a rich opportunity to become a leader in international trade and cater to this increasing demand by leveraging the country's abundant resources.⁴³



Contribute to the green jobs market:

The sector is expected to generate approximately six lakh jobs by 2030. The Ministry of New and Renewable Energy plans to implement several skill-building initiatives by collaborating with the Ministry of Education and the Ministry of Skill Development and Entrepreneurship to build a skilled workforce for India's green hydrogen transition.⁴⁴

³⁹ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

⁴⁰ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

⁴¹ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

⁴² Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

⁴³ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

⁴⁴ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

Challenges

This section highlights some of the major challenges and constraints facing green hydrogen in India. Efforts to resolve such challenges will ensure cost effective domestic manufacturing of GH2 components, development of demand clusters to boost adoption, state-level support for GH2 production and the deployment of a skilled GH2 workforce.



Inadequate resource assessments to create demand clusters:

Green hydrogen plants are most cost-effective in regions abundant in both solar and wind energy resources, such as Bhuj and Bhavnagar in Gujarat.⁴⁵ However, projections show that most future demand clusters housing key GH2-reliant industries such as fertilisers, refineries, and iron and steel only have access to solar resources.⁴⁶ India also needs to ramp up evaluation and analysis of various end-use sectors such as steel, ammonia, and refineries to determine which sectors are prepared for widespread adoption of green hydrogen and which sectors are suitable for conducting small- and large-scale pilot projects to test its feasibility and effectiveness.



Lack of domestic manufacturing of crucial components of hydrogen production, especially electrolyser:

India had six alkaline electrolyser manufacturers in 2020, far below the required capacity to make electrolyser technology efficient and economical.⁴⁷ As per the target set under the National Hydrogen Mission, of producing 5 MMT of green hydrogen per annum by 2030, India will need to increase its current electrolyser capacity to 60-100 GW.⁴⁸ Further, some public sector utilities (PSUs) in India can manufacture balance of plant (BoP) components, but they do not have the capacity to produce electrochemical stacks, which are critical components in the electrolysis process that facilitate the separation of hydrogen gas in the electrolysis process.⁴⁹



Cost and infrastructure challenges:

The levelised cost of green hydrogen production (LCOH) ranges from approximately INR 300 to INR 500 per kilogram, which is about double that of grey hydrogen production, priced at INR 140 to INR 190 per kilogram.⁵⁰ Further, low competitiveness of hydrogen-based products makes it difficult for it to compete with cheaper alternatives, hindering their uptake. For example, in the transport sector, fuel cell electric vehicles (FCEVs) have a higher ownership cost as compared to combustion engine vehicles. In industries, grey hydrogen or fossil fuels are preferred in applications like use as chemical feedstock or combustion fuel as they are significantly cheaper than green hydrogen.⁵¹



Storage, distribution, and transportation hurdles:

Storing hydrogen in pressurised steel tanks is expensive, as are the energy costs associated with chemical storage methods.⁵² Further, hydrogen transportation can be dangerous as it is highly flammable and prone to leakage. Setting up new transportation pipelines faces prohibitive high upfront costs. India can look at retrofitting existing pipelines, but it is only possible to an extent, due to the flammable nature of hydrogen and high costs. There is also

⁴⁵ CEEW, A Green Hydrogen Economy for India, 2020

⁴⁶ CEEW, A Green Hydrogen Economy for India, 2020

⁴⁷ NITI Aayog, Harnessing Green Hydrogen, 2022

⁴⁸ Ministry of New and Renewable Energy, 3-Day International Conference on Green Hydrogen begins in New Delhi, 2023

⁴⁹ NITI Aayog, Harnessing Green Hydrogen, 2022

⁵⁰ CEEW, Catalysing Green Hydrogen Growth in India, 2021

⁵¹ CEEW, Catalysing Green Hydrogen Growth in India, 2021

⁵² NITI Aayog, Harnessing Green Hydrogen, 2022

a lack of last-mile infrastructure to encourage uptake by end-consumers, like refuelling stations, which require substantial investments to establish.⁵³



Inadequate state-level focus on GH2:

Each state in India possesses a distinctive geographic and resource landscape that can define its green hydrogen production capabilities. However, not all states have tailored roadmaps leveraging their unique strengths and addressing their specific needs to meet national hydrogen targets effectively.⁵⁴ A few states have started creating their own GH2 policies, for instance, Uttar Pradesh has a draft green hydrogen policy, and the Gujarat government plans to release its first draft in 2023.⁵⁵ Odisha has also released its renewable energy policy which has provisions for GH2 production.⁵⁶ Further, currently, most GH2 R&D funding is provided by the central government. The National Green Hydrogen Mission emphasises the need for setting up Centres of Excellence to foster innovation in GH2 and related technology development that can promote industry, government, and academia collaboration at the state level.⁵⁷



Need to build GH2 specific skills in India's current energy workforce:

Due to the limited commercial production of GH2 in India, there is a shortage of skilled workers who possess the necessary expertise to work in various aspects of the GH2 value chain. These roles include plant operators, manufacturing technicians, computer-controlled machine operators, process safety technicians, hydrogen pipeline construction workers, and battery repair personnel.⁵⁸ The government has recently, in 2023, outlined steps to build these capacities via large scale educational and skilling programmes.⁵⁹

53 CEEW, Catalysing Green Hydrogen Growth in India, 2021; NITI Aayog, Harnessing Green Hydrogen, 2022; South Asia Regional Energy Partnership, Investment Landscape of Green Hydrogen in India, 2023

54 NITI Aayog, Harnessing Green Hydrogen, 2022

55 Uttar Pradesh New and Renewable Energy Development Agency, Uttar Pradesh Green Hydrogen Policy, 2022; Live Mint, Gujarat's draft of green hydrogen policy to be ready in 2 months, 2023

56 Government of Odisha, Odisha Renewable Energy Policy, 2022

57 Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

58 Skill Council for Green Jobs, Gearing up the Indian workforce for a Green Economy, 2023

59 Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee green hydrogen energy.



NITI Aayog

It conducts policy analysis of the green hydrogen energy sector; and undertakes steps to improve energy planning, and data management in India



Ministry of Power

The ministry administers, formulates, and implements power sector-related policies and provides R&D and technical assistance relating to grid integration, transmission system networks and distribution systems in the States/UTs



Ministry of New and Renewable Energy (MNRE)

The ministry develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power, transport and support country's objective of achieving energy security



Bureau of Energy Efficiency

The bureau is the nodal authority for all GH2 projects in India, assists in developing policies and strategies to promote energy efficiency and conservation and utilises existing resources and infrastructure in performing functions under the Energy Conservation Act



Indian Renewable Energy Development Agency

The agency finances and supports self-sustainable investments in energy generation from renewable sources, and carry out promotion of energy efficiency and sustainable environmental technologies

Major policies and programs

This section highlights existing policies related to green hydrogen energy and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



National Green Hydrogen Mission, 2023⁶⁰

The mission aims to position India as a global leader in the production, use and export of green hydrogen through the following key provisions:

- Ramp up domestic capacity to annually produce green hydrogen up to a minimum of 5 MMT by 2030 by robust R&D efforts, reforming regulations, and pilot programs

⁶⁰ Ministry of New and Renewable Energy, National Green Hydrogen Mission, 2023

- Develop and scale production-related technology, and build the infrastructure required for storage and delivery to make green hydrogen accessible and affordable



Green Hydrogen Standard for India, 2023⁶¹

The standards aim to provide regulatory clarity for upcoming green hydrogen projects by providing the following provisions:

- Outlining the amount of carbon emissions that are allowed in the process of producing green hydrogen from electrolysis and biomass
- Providing a clear definition of what encompasses green hydrogen and its production
- Notifying the Bureau of Energy Efficiency under the Ministry of Power as the nodal authority for GH2 projects

⁶¹ Ministry of New and Renewable Energy, Green Hydrogen Standard for India, 2023

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. Some ideas to enable progress given India's ambitious green hydrogen goals include:

 Enhance domestic manufacturing and production	- Consider financial incentives such as price decline targets and investment facilitation to help build India's GH2 producer base, promote domestic manufacturing of electrolyzers, and enable deployment of solar-wind hybrid plants for GH2 to accelerate private sector growth - Accelerate R&D in allied innovations and technologies, such as bioenergy for GH2 - Accelerate the upgradation of storage and distribution infrastructure
 Capacity building and skill development	- Develop skilling programmes for different roles (both for new entrants and to facilitate transitions from traditional energy sectors)
 Encourage adoption by various sectors	- Conduct assessments to identify sector-wise industrial hubs that can transition to using GH2, and use complementary policy instruments such as sectoral mandates for GH2 use and financial incentives for offtakers to accelerate demand - Provide state-level capacity-building support and technical assistance for policymakers to effectively plan, promote, and shift behaviours across end-use sectors

Recommended readings

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